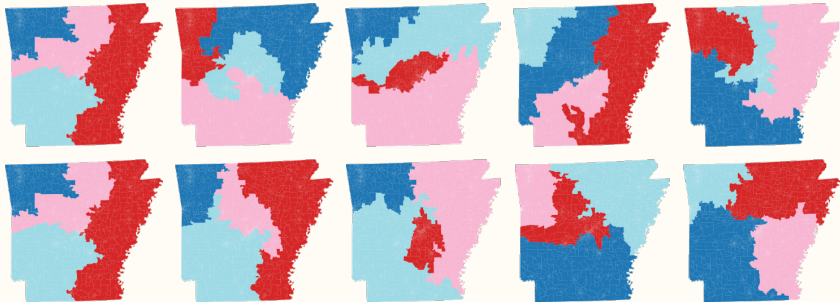


Bayesian Estimation of Block-Level Citizen Voting-Age Population for Redistricting

Christopher Wolfram, Moon Duchin, Aaron Schein

University of Chicago

Assessing districting plans



CVAP Sources

- Decennial Census?

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 - No **citizenship** information

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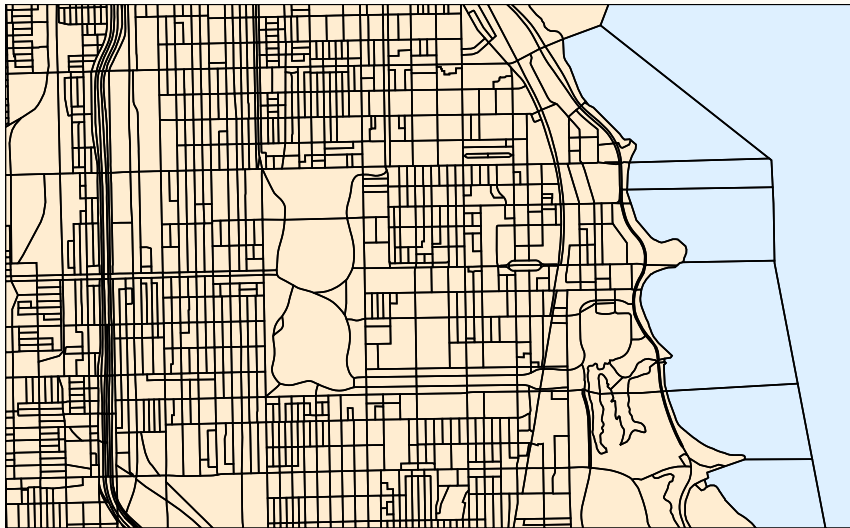
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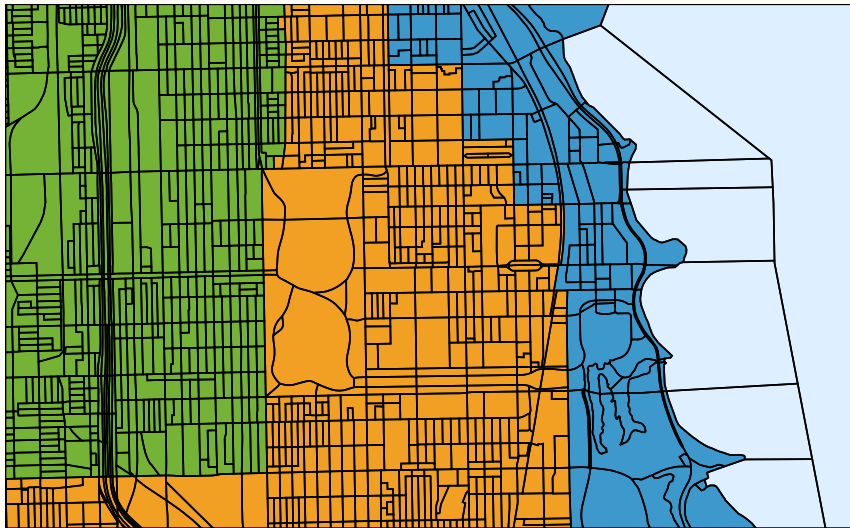
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Spatial Resolution & Small Area Estimation

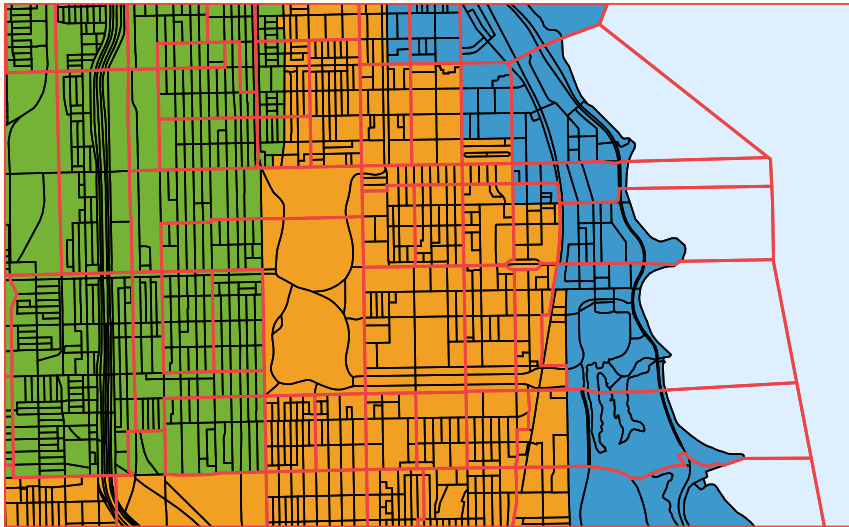
Redistricting Geography



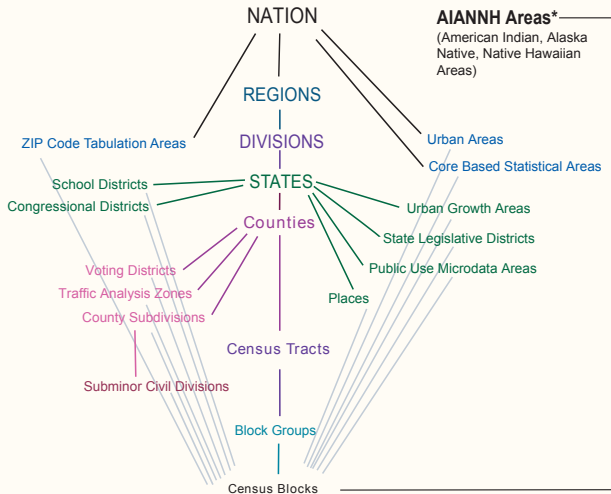
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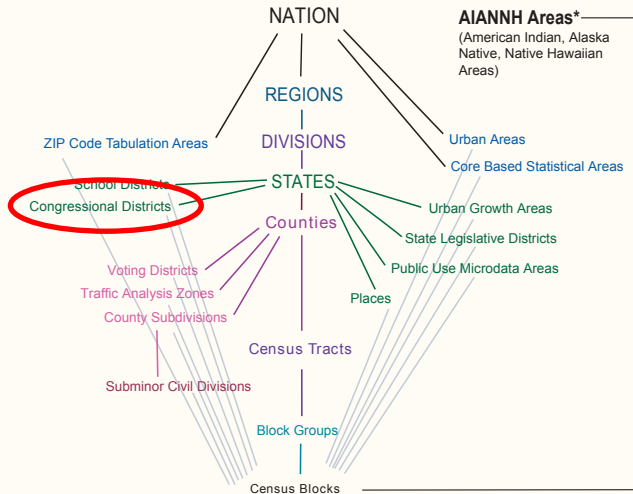
Redistricting Geography



Redistricting Geography



Redistricting Geography



Observed Variables

VAP^{PL}
block-level

$VAP^{ACS}, CVAP^{ACS}$
tract-level

Dissaggregation

$$\widehat{\text{CVAP}}_b = \text{CVAP}_{t[b]}^{\text{ACS}} \times \frac{\text{CVAP in } b}{\text{CVAP in } t[b]}$$

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Dissaggregation vs. Discounting

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What if $\text{VAP}_{t[b]}^{\text{ACS}} = 0$?

Practical Problem with Discounting

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What if $\text{VAP}_{t[b]}^{\text{ACS}} = 0$?

What if $\text{VAP}_{t[b]}^{\text{ACS}}$ is just small?

Why is this not a problem for Dissaggregation?

It is! It's not a numerical problem, but it is a statistical problem.

Bayesian Discounting

$$\omega_{t[b]} \sim \text{Beta}(\alpha, \beta)$$

$$\text{CVAP}_{t[b]}^{\text{ACS}} \sim \text{Binomial}(\text{VAP}_{t[b]}^{\text{ACS}}, \omega_{t[b]})$$

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- Discrete support!

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- Discrete support!
- Uncertainty quantification!

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- Discrete support!
- Uncertainty quantification!
- No numerical problems!

Beta-Binomial Conjugacy

Prior and likelihood:

$$\omega_{t[b]} \sim \text{Beta}(\alpha, \beta)$$

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Posterior:

$$\omega_{t[b]} \mid - \sim \text{Beta}\left(\alpha + \text{CVAP}_{t[b]}^{\text{ACS}}, \beta + \text{VAP}_{t[b]}^{\text{ACS}} - \text{CVAP}_{t[b]}^{\text{ACS}}\right)$$

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Posterior mean:

$$\hat{\omega}_{t[b]} = \frac{\alpha + \text{CVAP}_{t[b]}^{\text{ACS}}}{\alpha + \beta + \text{VAP}_{t[b]}^{\text{ACS}}}$$

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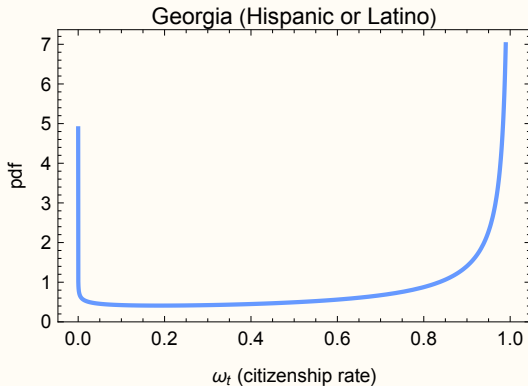
Posterior mean:

$$\hat{\omega}_{t[b]} = \frac{\alpha + \text{CVAP}_{t[b]}^{\text{ACS}}}{\alpha + \beta + \text{VAP}_{t[b]}^{\text{ACS}}} \approx \underbrace{\frac{\text{CVAP}_{t[b]}^{\text{ACS}}}{\text{VAP}_{t[b]}^{\text{ACS}}}}_{\text{naive estimator}}$$

Fitting the Bayesian Discounting Model

Citizenship rate prior:

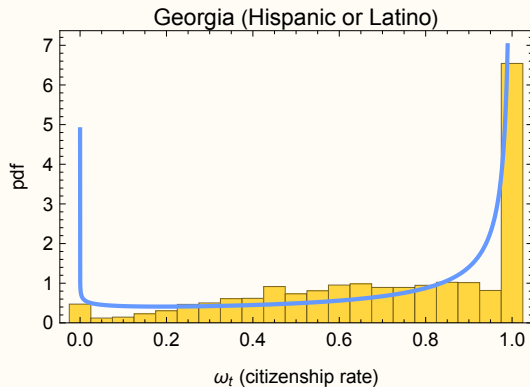
$$\omega_{t[b]} \sim \text{Beta}(\alpha, \beta), \quad \alpha \approx 0.84, \quad \beta \approx 0.29$$



Fitting the Bayesian Discounting Model

Citizenship rate prior:

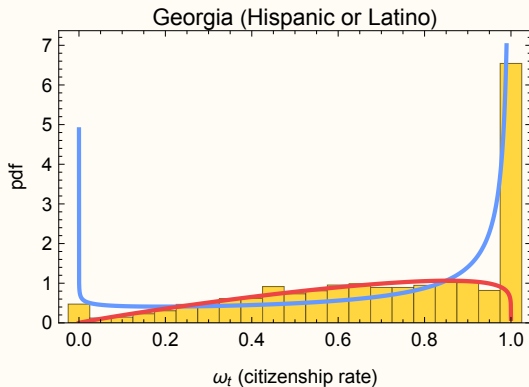
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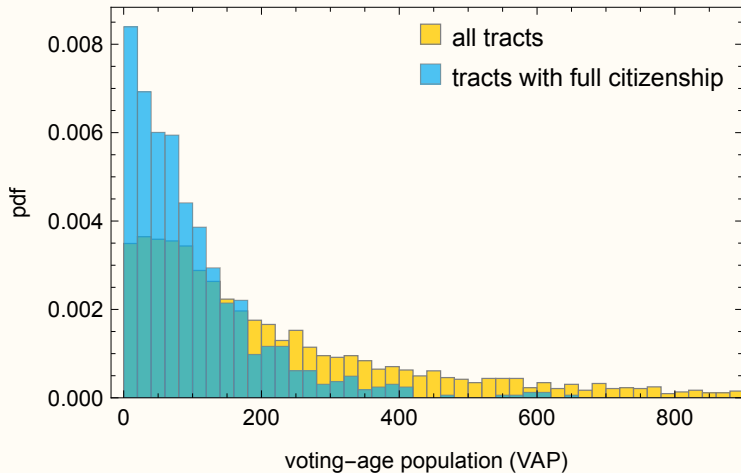
Fitting the Bayesian Discounting Model

Citizenship rate prior:

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It is not just tiny tracts



Bonus: Bayesian Dissaggregation

$$\omega_b \sim \text{Beta}(\alpha, \beta)$$

$$\text{VAP}_b^{\text{PL}} \sim \text{Binomial}(\text{VAP}_{t[b]}^{\text{PL}}, \omega_b)$$

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Race/Ethnicity Categories

Challenges

- The race/ethnicity categories used for ACS CVAP are not those relevant for litigation.
- The ACS race question differed from the Decennial Census question until 2020.

Census Questionnaire

6. Is this person of Hispanic, Latino, or Spanish origin?

- ☐ No, not of Hispanic, Latino, or Spanish origin
- ☐ Yes, Mexican, Mexican Am., Chicano
- ☐ Yes, Puerto Rican
- ☐ Yes, Cuban
- ☐ Yes, another Hispanic, Latino, or Spanish origin – *Print, for example, Salvadoran, Dominican, Colombian, Guatemalan, Spaniard, Ecuadorian, etc.* 

[illegible]

7. What is this person's race?

Mark ☒ one or more boxes **AND** print origins.

- White – Print, for example, German, Irish, English, Italian, Lebanese, Egyptian, etc. ↗

[illegible]

- Black or African Am. – *Print*, for example, African American, Jamaican, Haitian, Nigerian, Ethiopian, Somali, etc. ▾

[illegible]

- ☐ American Indian or Alaska Native – Print name of enrolled or principal tribe(s), for example, Navajo Nation, Blackfeet Tribe, Mayan, Aztec, Native Village of Barrow Inupiat Traditional Government, Nome Eskimo Community, etc.

[illegible]

- | | | |
|--|-------------------------------------|--|
| ☐ Chinese | ☐ Vietnamese | ☐ Native Hawaiian |
| ☐ Filipino | ☐ Korean | ☐ Samoan |
| ☐ Asian Indian | ☐ Japanese | ☐ Chamorro |
| ☐ Other Asian –
Print, for example
Pakistani, Cambodian,
Hmong, etc. ✓ | | ☐ Other Pacific Islander –
Print, for example,
Tongan, Fijian,
Marshallese, etc. ✓ |

[illegible]

- ☐ Some other race – *Print race or origin.* ↗

[illegible]

ACS Detailed Tables Race/Ethnicity Categories

- White Alone
- Black or African American Alone
- American Indian and Alaska Native Alone
- Asian Alone
- Native Hawaiian and Other Pacific Islander Alone
- Some Other Race Alone
- Two or More Races
- Hispanic or Latino
- White Alone, Not Hispanic or Latino
- Total

Let $\mathcal{X}_{g,h,r_W,r_B,\dots,r_O}$ be the count of people in geography g with Hispanic or Latino ethnicity h and race categories $r_W, r_B, \dots, r_O$

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For example,

$$\mathcal{X}_{g=1029,h=1,r_W=0,r_B=0,r_A=1,\dots,r_O=1} = 21$$

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For example,

$$\mathcal{X}_{g=1029,h=1,r_W=0,r_B=0,r_A=1,\dots,r_O=1} = 21$$

$$\mathcal{X} \in \mathbb{Z}_+^{|G| \times 2 \times 2 \dots}$$

$$\mathcal{X}_{g,r_W=1} = \sum_{h \in \{0,1\}} \mathcal{X}_{g,h,r_W=1,r_B=0,\dots,r_O=0} \quad (\text{White Alone})$$

$$\mathcal{X}_{g,r_B=1} = \sum_{h \in \{0,1\}} \mathcal{X}_{g,h,r_W=0,r_B=1,\dots,r_O=0} \quad (\text{Black or African American Alone})$$

$\vdots$

$$\mathcal{X}_{g,2+} = \sum_{h,r_W,r_B,\dots,r_O \in \{0,1\}} \mathbf{1}(r_W + r_B + \dots + r_O \geq 2) \mathcal{X}_{g,h,r_W,r_B,\dots,r_O} \quad (\text{Two or More})$$

$$\mathcal{X}_{g,h=1} = \sum_{r_W,r_B,\dots,r_O \in \{0,1\}} \mathcal{X}_{g,h=1,r_W,r_B,\dots,r_O} \quad (\text{Hispanic or Latino})$$

$$\mathcal{X}_{g,h=0,r_W=1} = \mathcal{X}_{g,h=0,r_W=1,r_B=0,\dots,r_O=0} \quad (\text{White Alone, Not Hispanic or Latino})$$

$$\mathcal{X}_g = \sum_{h,r_W,r_B,\dots,r_O \in \{0,1\}} \mathcal{X}_{g,h,r_W,r_B,\dots,r_O} \quad (\text{Overall})$$

$$\mathcal{X}_{g,h,r_W=0,r_B=0,\dots,r_O=0} = 0 \quad (\text{No race selected})$$

$$\mathcal{X}_{g^*,h,r_W,r_B,\dots,r_O} = \sum_{g \in g^*} \mathcal{X}_{g,h,r_W,r_B,\dots,r_O} \quad (\text{PUMS})$$

Equivalent Formulation

Let $x \in \mathbb{Z}_+^n$ be a flattened version of $\mathcal{X}$.

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Equivalent Formulation

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The problem: We observe $\tilde{y}$ and S_j and want to recover x .

Least Squares Formulation

$$y_j = \sum_{i \in S_j} x_i$$

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$$A_{j,i} = \mathbf{1}\{i \in S_j\}$$

$$y = Ax$$

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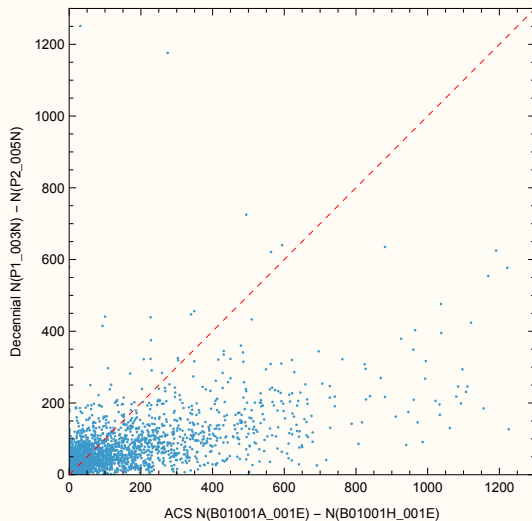
Minimize $\|Ax - \tilde{y}\|_2^2$ subject to $x \in \mathbb{Z}_+^n$

Poisson-Poisson Model

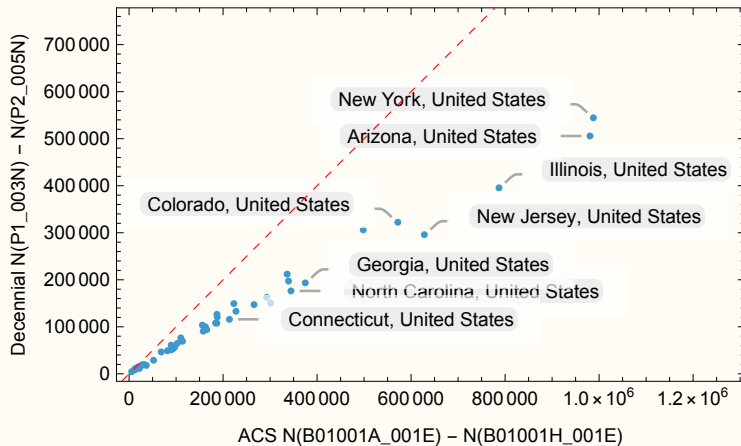
$$x_i \sim \text{Poisson}(\lambda_i)$$

$$\tilde{y}_j \sim \text{Poisson}\left(\sum_{i \in S_j} \lambda_i\right)$$

ACS vs. Decennial Census Differences



ACS vs. Decennial Census Differences



Questionnaire Changes

7. What is this person's race?

Mark ☒ one or more boxes **AND** print origins.

- ☐ White – Print, for example, German, Irish, English, Italian, Lebanese, Egyptian, etc.

- ☐ Black or African Am. – Print, for example, African American, Jamaican, Haitian, Nigerian, Ethiopian, Somali, etc.

- ☐ American Indian or Alaska Native – Print name of enrolled or principal tribe(s), for example, Navajo Nation, Blackfeet Tribe, Mayan, Aztec, Native Village of Barrow Inupiat Traditional Government, Nome Eskimo Community, etc.

- | | | |
|---|--|--|
| ☐ Chinese | ☐ Vietnamese | ☐ Native Hawaiian |
| ☐ Filipino | ☐ Korean | ☐ Samoan |
| ☐ Asian Indian | ☐ Japanese | ☐ Chamorro |
| ☐ Other Asian – Print, for example, Pakistani, Cambodian, Hmong, etc. <input type="text"/> | ☐ Other Pacific Islander – Print, for example, Tongan, Fijian, Marshallese, etc. <input type="text"/> | |

- ☐ Some other race – Print race or origin.

2020 Decennial Census

6 What is Person 1's race?

Mark (X) one or more boxes **AND** print origins.

- ☐ White – Print, for example, German, Irish, English, Italian, Lebanese, Egyptian, etc.

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2020 ACS

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- ☐ White
☐ Black or African Am.
☐ American Indian or Alaska Native – Print name of enrolled or principal tribe.

- | | | |
|---|--|--|
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| ☐ Chinese | ☐ Korean | ☐ Guamanian or Chamorro |
| ☐ Filipino | ☐ Vietnamese | ☐ Samoan |
| ☐ Other Asian – Print race, for example, Hmong, Laotian, Thai, Pakistani, Cambodian, and so on. <input type="text"/> | ☐ Other Pacific Islander – Print race, for example, Fijian, Tongan, and so on. <input type="text"/> | |

- ☐ Some other race – Print race.

2019 ACS

ACS vs. Decennial Census Differences

